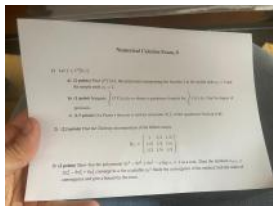


$x_0 = -1, x_1 = 1$ $L_0 f(x) = \sum_{i=0}^1 \mathcal{L}_i(x) f(x_i)$ $\mathcal{L}_i(x) = \prod_{j \neq i} \frac{(x - x_j)}{(x_i - x_j)}$
 $\mathcal{L}_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - 1}{-1 - 1} = \frac{1}{2}(1 - x)$
 $\mathcal{L}_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x + 1}{1 - (-1)} = \frac{1}{2}(1 + x)$

$L_m f(x) = \frac{1}{2}(1-x)f(-1) + \frac{1}{2}(1+x)f(1)$
 $= \frac{1}{2}(f(-1) - xf(-1) + f(1) + xf(1))$
 $= \frac{1}{2}(x(f(1) - f(-1)) + f(1) + f(-1))$
 $= \frac{1}{2}x(f(1) - f(-1)) + \frac{1}{2}(f(1) + f(-1))$

$b) \int_{-1}^1 (Pf)(x) dx > \int_{-1}^1 f(x) dx$
 $\int_{-1}^1 L_m f(x) dx = \frac{1}{6}(f(1) - f(-1))x^2 + \frac{1}{2}(f(-1) + f(1))x \Big|_{-1}^1$
 $= \frac{1}{6}(f(1) - f(-1)) \cdot 1^2 + \frac{1}{2}(f(-1) + f(1)) \cdot 1 -$
 $-\frac{1}{6}(f(1) - f(-1)) \cdot (-1)^2 - \frac{1}{2}(f(-1) + f(1)) \cdot (-1)$
 $= f(-1) + f(1)$



$L_m f(x) = \int_a^b K_m(x, t) f(t) dt$
 $K_m(x, t) = \frac{1}{m!} t^m (x - t)^{m-1}$

$L_m f = \frac{1}{(m+1)!} f^{(m+1)}(\xi) L_m(x)$
 $L_m f = \frac{1}{(m+1)!} f^{(m+1)}(\xi) L_m(x)$

$\int_a^b w(x) f(x) dx = \sum_{i=0}^m \lambda_i F_i(f) + M(f)$
 $M(f) = \frac{1}{(m+1)!} f^{(m+1)}(\xi) \int_a^b w(x) L_m(x) dx$

Hermite Interpolation

double nodes $H_m f(x) = \sum_{i=0}^m [h_{0i}(x)f(x_i) + h_{1i}(x)f'(x_i)]$ (1.3)

$h_{0i}(x) = [1 - 2l_i(x)]l_i(x) \prod_{j \neq i} [l_j(x)]^2$ (1.4)
 $h_{1i}(x) = (x - x_i)l_i(x) \prod_{j \neq i} [l_j(x)]^2$, $i = 0, \dots, m$

Newton's divided difference form

$N_m(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)^2 + \dots$
 $+ f[x_0, x_1, x_2, x_3](x - x_0)^3 + \dots + f[x_0, x_1, x_2, x_3, x_4](x - x_0)^4 + \dots$ (1.6)
 $h_{0i}(x) = f(x) - H_m f(x) = [n, x] \prod_{j=0}^{m-1} \frac{(x - x_j)^2}{(x_i - x_j)^2}$, $x \in (a, b)$ (1.9)
 $h_{0i}(x) = \frac{(x - x_i)^2}{f'} \left[\sum_{j=0}^{m-1} \frac{(x - x_j)^2}{(x_i - x_j)^2} \right] h_{0i}(x)$ (1.13)
 $N_m(x) = f(x_0) + f[x_0, x_1](x - x_0) + \dots$ (1.14)
 $R_m(x) = \frac{w(x)}{(m+1)!} f^{(m+1)}(\xi)$, $\xi \in (a, b)$

1) Let $f \in C^2[0, 1]$.

a) (2 points) Find $(Pf)(x)$, the polynomial interpolating the function f at the double node $x_0 = 0$ and the simple node $x_1 = 1$.

b) (3 points) Integrate $\int_0^1 (Pf)(x) dx$ to obtain a quadrature formula for $\int_0^1 f(x) dx$. Find its degree of precision.

c) (1.5 points) Use Peano's theorem to find the remainder $R(f)$ of the quadrature formula in b).

$x_0 = 0$ double node $\Rightarrow$ stem $f(0), f'(0)$ $\gamma_0 = \{0, 1\}$
 $x_1 = 1$ single node $\Rightarrow$ stem $f(1)$ $\gamma_1 = \{0\}$
 $n = 1 + 1 + 1 - 1 = 2$ = nr. de informatii pe care le avem
 $\{f(0), f'(0), f(1)\} - 1$

z	$f(z)$	$\Delta^1 f$	$\Delta^2 f$
$z_0 = x_0$	$f(0)$	$\frac{f(0) - f'(0)}{x_0 - x_0} = f'(0)$	$\frac{f(1) - f(0) - f'(0)}{x_1 - x_0} = \frac{f(1) - f(0) - f'(0)}{1}$
$z_1 = x_0$	$f'(0)$	$\frac{f(1) - f(0) - f'(0)}{x_1 - x_0} = \frac{f(1) - f(0) - f'(0)}{1}$	$\frac{f(1) - f(0) - f'(0)}{x_1 - x_0} = \frac{f(1) - f(0) - f'(0)}{1}$
$z_2 = x_1$	$f(1)$		

Newton's divided difference form

$N_m(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)^2 + \dots$
 $+ f[x_0, x_1, x_2, x_3](x - x_0)^3 + \dots + f[x_0, x_1, x_2, x_3, x_4](x - x_0)^4 + \dots$ (1.6)

$N_2(x) = f(0) + f'(0)(x - x_0) + (f(1) - f(0) - f'(0))(x - x_0)^2$
 $= f(0) + f'(0)x + (f(1) - f(0) - f'(0))x^2$
 $b) \int_0^1 f(0) + \int_0^1 f'(0)x + \int_0^1 (f(1) - f(0) - f'(0))x^2 =$

$$b) \int_0^1 f(x) dx \approx \int_0^1 f'(x) dx + (f(1) - f(0) - f'(0))x =$$

$$f(0) \cdot x \Big|_0^1 + f'(0) \frac{x^2}{2} \Big|_0^1 + (f(1) - f(0) - f'(0)) \frac{x^3}{3} \Big|_0^1$$

$$= f(0)(1-0) + f'(0)(\frac{1}{2} - 0) + (f(1) - f(0) - f'(0))(\frac{1}{3} - \frac{0}{3})$$

$$= f(0) + \frac{1}{2}f'(0) + \frac{1}{3}(f(1) - f(0) - f'(0))$$

$$= \cancel{f(0)} - \cancel{\frac{f(0)}{3}} + \cancel{\frac{f'(0)}{2}} - \cancel{\frac{f'(0)}{3}} + \frac{f(1)}{3} = \frac{2f(0)}{3} + \frac{f'(0)}{6} + \frac{f(1)}{3}$$

$$= \frac{2f(0) + f(1)}{3} + \frac{f'(0)}{6} = \frac{4f(0) + 2f(1) + f'(0)}{6}$$

$$R(f) = \int_0^1 f(x) dx - \int_0^1 (Pf)(x) dx$$

$$\Rightarrow R(f) = \int_0^1 f(x) dx - \frac{4f(0) + 2f(1) + f'(0)}{6}$$

$$f(x) = e_i(x) = x^i$$

$$R(e_0(x)) = \int_0^1 x^0 dx - \frac{4e_0(0) + 2e_0(1) + e_0'(0)}{6}$$

$$= x \Big|_0^1 - \frac{4+2+0}{6} = 1 - 0 - 1 = 0$$

$$R(e_1(x)) = \int_0^1 x^1 dx - \frac{4e_1(0) + 2e_1(1) + e_1'(0)}{6}$$

$$= \frac{x^2}{2} \Big|_0^1 - \frac{0+2+1}{6} = \frac{1}{2} - 0 - \frac{3}{6} = \frac{1}{2} - \frac{1}{2} = 0$$

$$R(e_2(x)) = \int_0^1 x^2 dx - \frac{4e_2(0) + 2e_2(1) - e_2'(0)}{6}$$

$$= \frac{x^3}{3} \Big|_0^1 - \frac{0+2-0}{6} = \frac{1}{3} - \frac{2}{6} = 0$$

$$R(e_3(x)) = \int_0^1 x^3 dx - \frac{4e_3(0) + 2e_3(1) - e_3'(0)}{6}$$

$$\frac{x^4}{4} \Big|_0^1 - \frac{0+2-0}{6} = \frac{1}{4} - \frac{2}{6} = \frac{1}{4} - \frac{1}{3} \neq 0$$

$$= \frac{3-4}{12} = -\frac{1}{12}$$

degree of precision = 2

$$c) f(x) = H_1(f(x)) + R(f(x)) \Rightarrow R(f(x)) = f(x) - H_2(f(x))$$

$$\int_0^1 (Pf)(x) dx + R(f)$$

quadrature formula

$$c) f(x) = H_2(f(x)) + R(f(x)) \Rightarrow R(f(x)) = f(x) - H_2(f(x))$$

$$Lf(x) = \int_a^x K_n(x,t) f^{(n+1)}(t) dt,$$

2) using Peano's Theorem for our test function $\Rightarrow$

$$K_n(x,t) = \frac{1}{n!} L((x-t)_+^n)$$

$$Lf = \frac{1}{(n+1)!} f^{(n+1)}(\xi) L e_{n+1},$$

$$\int_a^b w(x) f(x) dx = \sum_{j=0}^m A_j f_j(f) + R(f),$$

$$\Rightarrow R(f(x)) = \int_0^1 K_2(x,t) f'''(t) dt$$

$$K_2(x,t) = \frac{1}{2!} R((x-t)_+^2)$$

$$H_2^{(x)} = f(0) + f'(0)x + \frac{1}{2!} (f''(0) - f'(0) - f'(0))x^2$$

$$R((x-t)_+^2) = (x-t)_+^2 - \underbrace{(0-t)_+^2}_{=0} + \frac{\partial}{\partial x} (x-t)_+^2 \Big|_{x=0} \cdot x + \left((1-t)_+^2 - \underbrace{(0-t)_+^2}_{=0} - \frac{\partial}{\partial x} (x-t)_+^2 \Big|_{x=0} \right) x^2$$

$$= (x-t)_+^2 + (2(x-t)_+ \Big|_{x=0}) \cdot x + ((1-t)^2 - 0 - 2(x-t)_+ \Big|_{x=0}) x^2$$

$$= (x-t)_+^2 + (2 \underbrace{(0-t)_+}_{=0}) \cdot x + ((1-t)^2 - 2 \underbrace{(0-t)_+}_{=0}) x^2$$

$$= (x-t)_+^2 + 0 + ((1-t)^2 - 0) x^2$$

$$= (x-t)_+^2 + x^2(1-t)^2 > 0$$

$$0 \leq x \leq t \leq 1 \Rightarrow (x-t)_+ = 0$$

$$\hookrightarrow 0^2 + x^2(1-t)^2 > 0$$

$$0 \leq t \leq x \leq 1 \Rightarrow (x-t)_+ = x-t$$

$$\hookrightarrow (x-t)^2 + x^2(1-t)^2 > 0$$

$$\int x^2$$

$\Rightarrow K$ has constant sign on $[0, 1] \Rightarrow$

corollary Peano $\Rightarrow Rf = \frac{1}{3!} f^{(3)}(\xi) \cdot Re_3$

$$\begin{aligned} R_2 f(x) &= \int_0^1 \frac{1}{3!} f^{(3)}(\xi) \cdot Re_3 = \frac{1}{6} f^{(3)}(\xi) \int_0^1 Re_3 \\ &= \frac{1}{6} f^{(3)}(\xi) \cdot R_2 e_3(x) = \frac{1}{6} f^{(3)}(\xi) \cdot \left(-\frac{1}{12}\right) \\ &= -\frac{1}{72} f^{(3)}(\xi) \end{aligned}$$

④ $f: [0, 1] \rightarrow \mathbb{R}$ with continuous 3rd order derivatives
a) Find quadrature formula w. max. degree of precision,
of the form: $\int_0^1 f(x) dx = A \cdot f(0) + B f'(x_1) + R(f)$
b) $f(0), f'(1/3) \Rightarrow Pf(x)$? Birkhoff
c) $\int_0^1 (Pf)(x) dx$
d) $R(f)$

$$a) R(f) = \int_0^1 f(x) dx - A \cdot f(0) - B f'(x_1)$$

$$\begin{aligned} R(e_0) &= \int_0^1 e_0(x) dx - A \cdot e_0(0) - B \cdot e_0'(x_1) \\ &= x \Big|_0^1 - A \cdot 1 - B \cdot 0 \end{aligned}$$

$$1 - A = 0 \quad \Rightarrow \quad A = 1$$

$$R(e_1) = \int_0^1 e_1(x) dx - 1 \cdot e_1(0) - B \cdot e_1'(x_1)$$

$$R(e_1) = \frac{x^2}{2} \Big|_0^1 - 1 - B = \frac{1}{2} - 1 - B$$

x,)

$$R(e_1) = \left. \frac{x^2}{2} \right|_0^1 - 0 - B = \frac{1}{2} - B$$

$$\frac{1}{2} - B = 0 \Rightarrow B = \frac{1}{2}$$

$$R(e_2) = \int_0^1 e_2(x) dx - 1 \cdot e_2(0) - \frac{1}{2} e_2'(0)$$

$$= \left. \frac{x^3}{3} \right|_0^1 - 0 - x_1 = \frac{1}{3} - x_1 \Rightarrow \frac{1}{3} - x_1 = 0$$

$$\Rightarrow x_1 = \frac{1}{3}$$

$$R(e_3) = \int_0^1 e_3(x) dx - 1 \cdot e_3(0) - \frac{1}{2} e_3'(0)$$

$$= \left. \frac{x^4}{4} \right|_0^1 - 0 - \frac{1}{2} \cdot \frac{3}{9} = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

degree of maximum precision is 2

$$\text{for } A = 1, B = \frac{1}{2}, x_1 = \frac{1}{3}$$

$$b) f(0), f'(\frac{1}{3}) \Rightarrow P_f(x)?$$

$$J_0 = \{0\}$$

$$\begin{matrix} x_0 = 0 \\ x_1 = \frac{1}{3} \end{matrix}$$

$$\Rightarrow n = 1 + 1 - 1_m = 1$$

$$J_1 = \{1\}$$

$$B_1 f(x) = \sum_{k=0}^n \sum_{j \in J_k} b_{kj} l_{kj}(x)$$

$x_1)$

$\neq 0$

$\frac{1}{3})$

$\neq 0$

$\neq 0$
 x_k

$$B_1 f(x) = b_{00}(x) f(x_0) + b_{11}(x) f'(x_0)$$

$$b_{00}(x) = ax + b$$

$$b_{00}'(x) = a$$

$$\begin{cases} b_{00}(x_0) = 1 \\ b_{00}'(x_1) = 0 \end{cases} \quad \Leftrightarrow \quad \begin{cases} a \cdot 0 + b = 1 \\ a = 0 \end{cases}$$

$$\begin{cases} b_{11}(x_0) = 0 \\ b_{11}'(x_1) = 1 \end{cases} \quad \Leftrightarrow \quad \begin{cases} a \cdot 0 + b = 0 \\ a = 1 \end{cases}$$

$$B_1 f(x) = 1 \cdot f(x_0) + x \cdot f'(x_0)$$

$$= f(0) + x \cdot f'(0)$$

$$c) \int_0^1 f(0) + x f'(\frac{1}{3}) dx = f(0)$$

$$= f(0) + \frac{1}{2} f'(\frac{1}{3})$$

1)

$$\left\{ \begin{array}{l} a = 0 \\ b = 1 \end{array} \right. \Rightarrow b_{00}(x) = 1$$

$$\left\{ \begin{array}{l} a = 1 \\ b = 0 \end{array} \right. \Rightarrow b_{11}(x) = x$$

$$f'(x_1)$$

$$f'\left(\frac{1}{3}\right)$$

$$f'(x) \Big|_0 + f'\left(\frac{1}{3}\right) \cdot \frac{x^2}{2} \Big|_0$$

$$U^{(1)} T \approx f^{(1)}$$

$$d) \quad f(x) \approx \cancel{B_1} f(x) + R_1 f(x)$$

$$R_1 f(x) = f(x) - B_1 f(x)$$

$$Rf = \int_0^1 f(x) dx - \int_0^1 B_1 f(x) dx$$

$$Rf = \int_0^1 f(x) dx - f(0) +$$

$$R_1 f(x) = f(x) - B_1 f(x)$$

$$R_1(f(x)) = \int_0^1 K_2(x, t) f'(t) dt$$

$$K_2(x, t) = \frac{1}{2!} R_1((x-t)_+^2)$$

$$R_1((x-t)_+^2) = (x-t)_+^2 - \underbrace{(0-t)_+^2}_{=0}$$

$$B_1(f(x)) = f(0) + x \cdot f'(\frac{1}{3})$$

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$$\left. \begin{array}{l} \int dx \\ \frac{1}{2} f'(\frac{1}{3}) \end{array} \right\} \text{fără asta}$$

$$\int (t) dt$$

$$\left. \begin{array}{l} \frac{1}{2} \\ \frac{1}{2} \end{array} \right) + X \frac{\partial}{\partial x} (x-t)^2 \Big|_{x=\frac{1}{3}}$$

$$B_1(f(x)) = f(0) + \dots + \frac{f^{(3)}(\xi)}{3!}$$

$$\begin{aligned} \mathbb{I} R_1((x-t)_+^2) &= (x-t)_+^2 - 0 - \frac{f'(0)}{1!} (x-t)_+^2 \\ &= \underbrace{(x-t)_+^2}_{\geq 0} + \underbrace{2x \left(\frac{1}{3} - \frac{1}{6}\right)}_{\geq 0} \end{aligned}$$

$$\Rightarrow Rf = \frac{1}{2!} f''(\xi)$$

$$\begin{aligned} \int_0^1 Rf &= \int_0^1 \frac{1}{2} f''(\xi) \\ &= \frac{1}{2} f''(\xi) \end{aligned}$$

$$+ x \cdot 2(x-t) + \left| x = \frac{1}{3} \right.$$

$t) + \Rightarrow$ Peano theorem
 ≥ 0

Re_2

$$) Re_3 \geq \frac{1}{2} f^{(4)}(\xi) \int_0^1 Re_3$$

$$, \frac{1}{12} \geq \frac{1}{24} f^{(4)}(\xi)$$